

## 5. Acquisition A: noise-plate series

This series captures the 9 Z stations of the ladder (Z\_STATION\_RATIO), plus the two LEGACY\_METRIC\_DEPTHS\_MM stations,  $\times 5$  field positions  $\times$  FRAMES\_PER\_NOISE\_STATION frames, plus a tilt sub-series. The station order is randomized so that slow drift cannot masquerade as a Z dependence.

1. Mount T2. Build the station list from every station of the Z ladder (Z\_STATION\_RATIO, Z\_MIN to Z\_MAX) and the LEGACY\_METRIC\_DEPTHS\_MM stations, at five field positions, all fronto-parallel. The positions are center, then  $(\pm, \pm)$  FIELD\_OFFSET\_FRACTION of the half-FOV. Check that the plate fully covers the analysis region of interest at every station. At Z\_MIN off-axis this may limit the field offset.
2. Shuffle the station list with a logged random seed.
3. Before the first station, capture a drift sentinel: center, Z\_REFERENCE\_MM, SENTINEL\_FRAMES frames. Repeat the sentinel every DRIFT\_SENTINEL\_INTERVAL\_MIN and after the last station.
4. At each station: move, wait ROBOT\_SETTLE\_TIME\_S, then capture FRAMES\_PER\_NOISE\_STATION consecutive frames. Log the read-back robot pose, the sensor temperature, and timestamps.
5. Tilt sub-series. At the center and at the reduced stations (Z\_REDUCED\_STATION\_STRIDE), tilt T2 about V, then about H, through each angle in TILT\_ANGLES\_DEG. Capture FRAMES\_PER\_TILT\_POSE frames per pose. Incidence angle changes both the per-pixel noise and the fill rate.
6. Repeat-mount check. Dismount T2, re-mount it, and repeat the Z\_REFERENCE\_MM center station. The difference shows how much of the bias comes from re-mounting the target.
7. Jitter-span check. Before any B, C, or D capture, run the autocorrelation quick-look of Section 10, Step 7 on one center station. If the correlation length exceeds PHASE\_JITTER\_SPAN\_PX, raise the span (Section 2) before continuing.
8. If Step 4.2 calls for it, repeat Steps 1–5 with the sensor's filters off.